title: “Democracia y Gasto Tributario Empresarial” author: “Frank” date: “2026-03-31” output: html_document: df_print: paged

0. Instalación y carga de paquetes

```
install.packages("here")
install.packages("readr")
install.packages("readxl")
install.packages("dplyr")
install.packages("stargazer")
install.packages("moments")
install.packages("fBasics")
install.packages("car")
install.packages("lmtest")
install.packages("tseries")
install.packages("sandwich")
install.packages("ggplot2")
install.packages("corrplot")
```

```
rm(list = ls())

library(here)        # Directorio de trabajo
library(readr)       # Leer CsV
library(readxl)      # Leer Excel (GTED)
library(dplyr)       # Manipulación de datos
library(stargazer)   # Tablas de resultados
library(moments)     # skewness y kurtosis
library(fBasics)     # Estadísticos básicos
library(car)         # Contrastes (VIF, REsET, etc.)
library(lmtest)      # Contrastes (BP, White, GQ)
library(tseries)     # Contraste Jarque-Bera
library(sandwich)    # Errores estándar robustos
library(ggplot2)     # Gráficos
library(corrplot)    # Matriz de correlaciones visual
```

1. Carga y nálisis exploratorio de datos

1.1 Estadísticos descriptivos

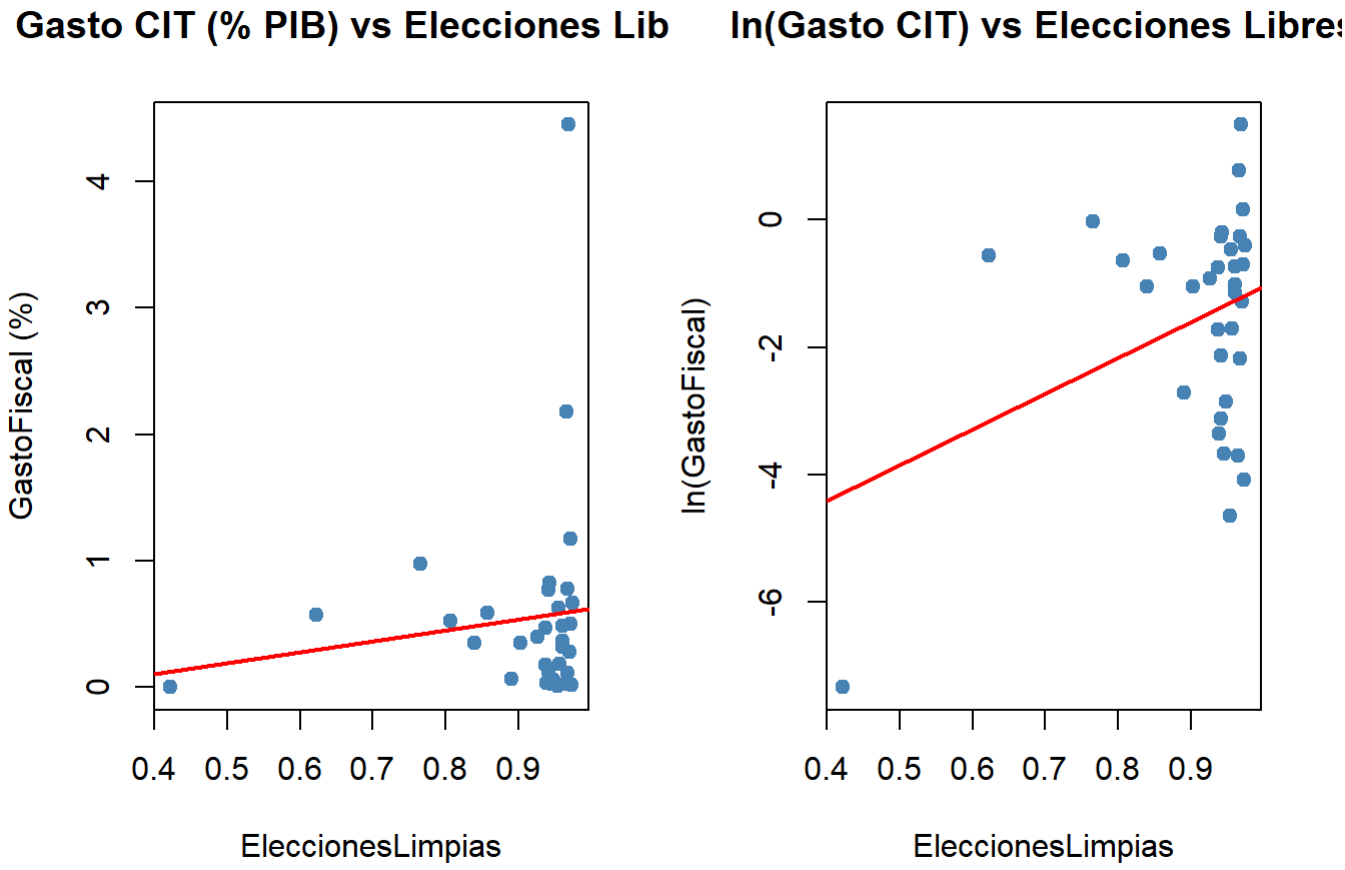
```
# Cargar el data frame desde el CsV
df <- read_csv("df_tfm_cs.csv", show_col_types = FALSE)
```

```
#  Nombres de variables
df <- df %>% rename(
  GastoFiscal        = te_cit_gdp,
  GastoFiscal_ln     = ln_te_cit,
  EleccionesLimpias  = v2xel_frefair,
  Ig_Distri_Recursos = v2xeg_eqdr,
  P_Inmi             = v2paimmig,
  t_Fecund           = e_miferrat,
  Populismo          = v2xpa_popul
)

# Actualizar los strings para que apunten a los nuevos nombres
GastoFiscal        <- "GastoFiscal"
GastoFiscal_ln     <- "GastoFiscal_ln"
EleccionesLimpias  <- "EleccionesLimpias"
ln_gdp             <- "ln_gdp"
Ig_Distri_Recursos <- "Ig_Distri_Recursos"
P_Inmi             <- "P_Inmi"
t_Fecund           <- "t_Fecund"
Populismo          <- "Populismo"

df.des <- df[, c(GastoFiscal, GastoFiscal_ln, EleccionesLimpias, ln_gdp,
                 Ig_Distri_Recursos, P_Inmi, t_Fecund, Populismo)]

media      <- sapply(df.des, mean,     na.rm = TRUE)
mediana    <- sapply(df.des, median,   na.rm = TRUE)
máximo     <- sapply(df.des, max,      na.rm = TRUE)
mínimo     <- sapply(df.des, min,      na.rm = TRUE)
desviación <- sapply(df.des, sd,       na.rm = TRUE)
asimetría  <- sapply(df.des, skewness, na.rm = TRUE)
curtosis   <- sapply(df.des, kurtosis, na.rm = TRUE)

round(rbind(media, mediana, máximo, mínimo, desviación, asimetría, curtosis), 3)
```

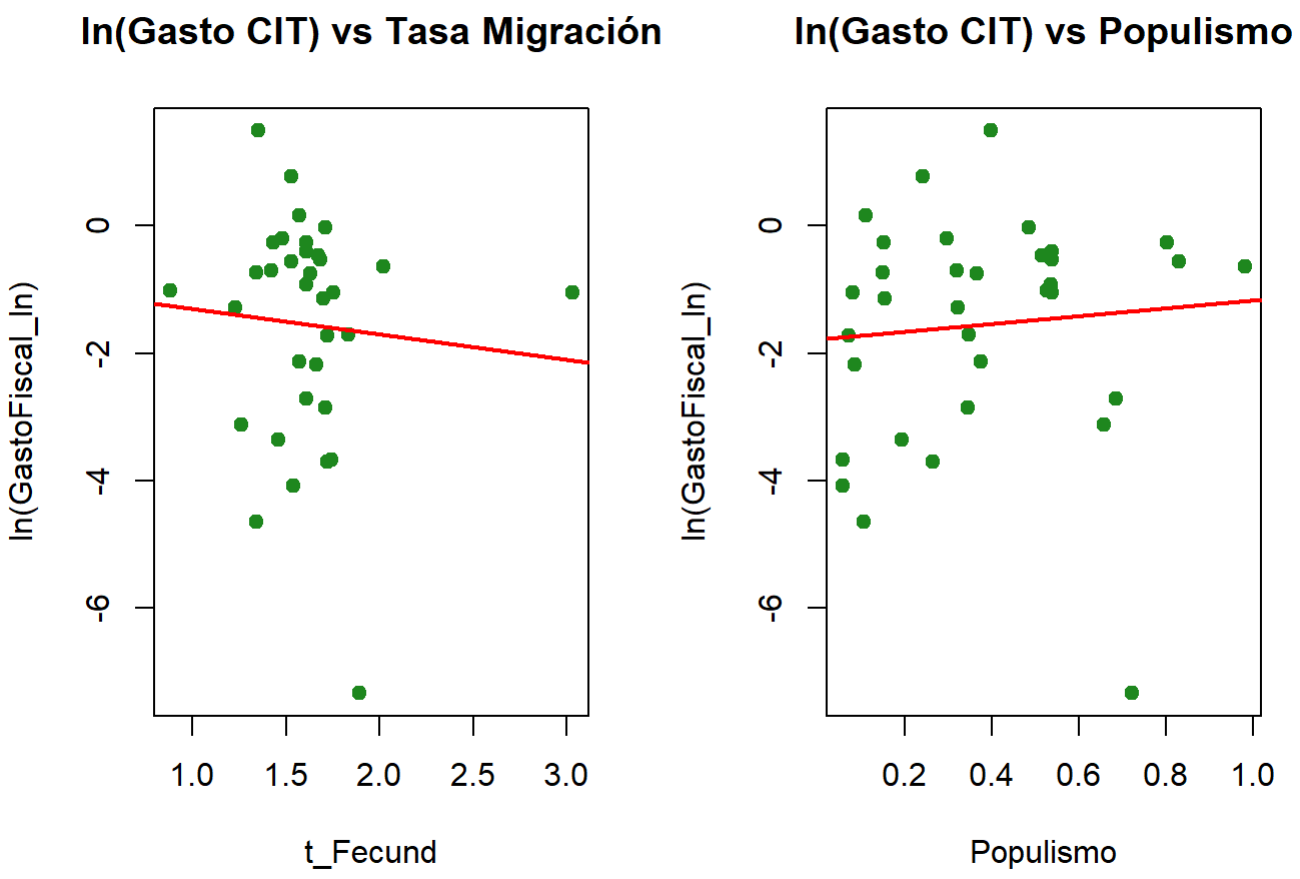
```
##            GastoFiscal GastoFiscal_ln EleccionesLimpias ln_gdp
## media            0.545         -1.551             0.910 26.712
## mediana          0.357         -1.030             0.946 26.633
## máximo           4.450          1.493             0.974 30.630
## mínimo           0.001         -7.340             0.422 23.748
## desviación       0.816          1.759             0.113  1.564
## asimetría        3.394         -1.129            -2.917  0.091
## curtosis        12.937          1.501             8.762 -0.329
##            Ig_Distri_Recursos P_Inmi t_Fecund Populismo
## media                   0.848 -0.183    1.613     0.377
## mediana                 0.907  0.494    1.610     0.345
## máximo                  0.984  2.989    3.030     0.980
## mínimo                  0.329 -4.797    0.880     0.057
## desviación              0.166  1.971    0.328     0.249
## asimetría              -1.793 -0.605    1.985     0.519
## curtosis                2.333 -0.524    8.205    -0.674
```

1.2 Gráficos de dispersión

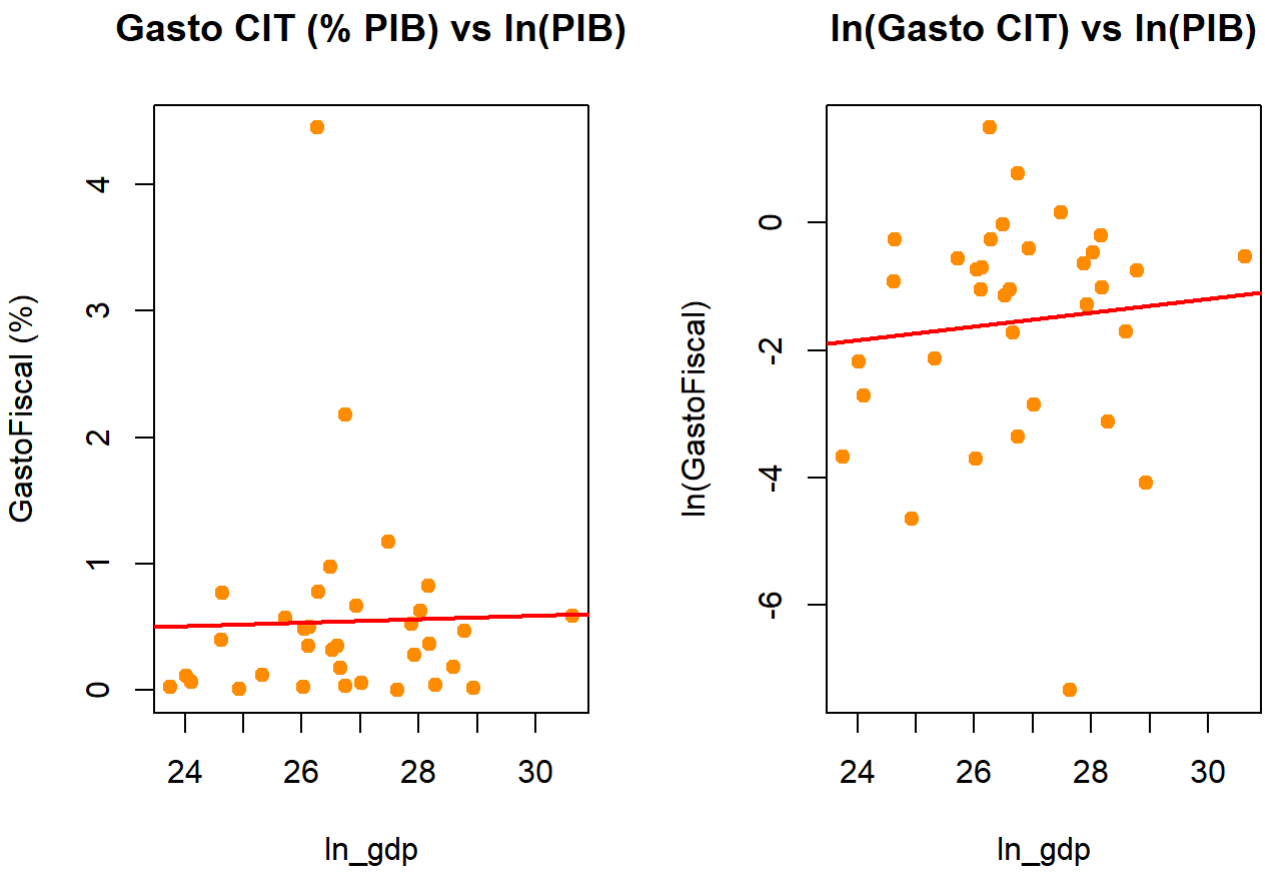
```
par(mfrow = c(1, 2))

plot(df[[GastoFiscal]] ~ df[[EleccionesLimpias]],
     main = "Gasto CIT (% PIB) vs Elecciones Libres",
     xlab = EleccionesLimpias,
     ylab = paste0(GastoFiscal, " (%)"),
     pch  = 19, col = "steelblue")
abline(lm(df[[GastoFiscal]] ~ df[[EleccionesLimpias]]), col = "red", lwd = 2)
text(df[[EleccionesLimpias]], df[[GastoFiscal]], labels = df$etiqueta, cex = 0.6, pos = 3)

plot(df[[GastoFiscal_ln]] ~ df[[EleccionesLimpias]],
     main = "ln(Gasto CIT) vs Elecciones Libres",
     xlab = EleccionesLimpias,
     ylab = paste0("ln(", GastoFiscal, ")"),
     pch  = 19, col = "steelblue")
abline(lm(df[[GastoFiscal_ln]] ~ df[[EleccionesLimpias]]), col = "red", lwd = 2)
text(df[[EleccionesLimpias]], df[[GastoFiscal_ln]], labels = df$etiqueta, cex = 0.6, pos = 3)
```



```
par(mfrow = c(1, 2))

plot(df[[GastoFiscal_ln]] ~ df[[t_Fecund]],
     main = "ln(Gasto CIT) vs Tasa Migración",
     xlab = t_Fecund,
     ylab = paste0("ln(", GastoFiscal_ln, ")"),
     pch  = 19, col = "forestgreen")
abline(lm(df[[GastoFiscal_ln]] ~ df[[t_Fecund]]), col = "red", lwd = 2)
text(df[[t_Fecund]], df[[GastoFiscal_ln]], labels = df$etiqueta, cex = 0.6, pos = 3)

plot(df[[GastoFiscal_ln]] ~ df[[Populismo]],
     main = "ln(Gasto CIT) vs Populismo",
     xlab = Populismo,
     ylab = paste0("ln(", GastoFiscal_ln, ")"),
     pch  = 19, col = "forestgreen")
abline(lm(df[[GastoFiscal_ln]] ~ df[[Populismo]]), col = "red", lwd = 2)
text(df[[Populismo]], df[[GastoFiscal_ln]], labels = df$etiqueta, cex = 0.6, pos = 3)
```



```
#  Nombres de variables (modificar aquí si cambian)

par(mfrow = c(1, 2))

plot(df[[GastoFiscal]] ~ df[[ln_gdp]],
     main = "Gasto CIT (% PIB) vs ln(PIB)",
     xlab = ln_gdp,
     ylab = paste0(GastoFiscal, " (%)"),
     pch  = 19, col = "darkorange")
abline(lm(df[[GastoFiscal]] ~ df[[ln_gdp]]), col = "red", lwd = 2)
text(df[[ln_gdp]], df[[GastoFiscal]], labels = df$etiqueta, cex = 0.6, pos = 3)

plot(df[[GastoFiscal_ln]] ~ df[[ln_gdp]],
     main = "ln(Gasto CIT) vs ln(PIB)",
     xlab = ln_gdp,
     ylab = paste0("ln(", GastoFiscal, ")"),
     pch  = 19, col = "darkorange")
abline(lm(df[[GastoFiscal_ln]] ~ df[[ln_gdp]]), col = "red", lwd = 2)
text(df[[ln_gdp]], df[[GastoFiscal_ln]], labels = df$etiqueta, cex = 0.6, pos = 3)
```

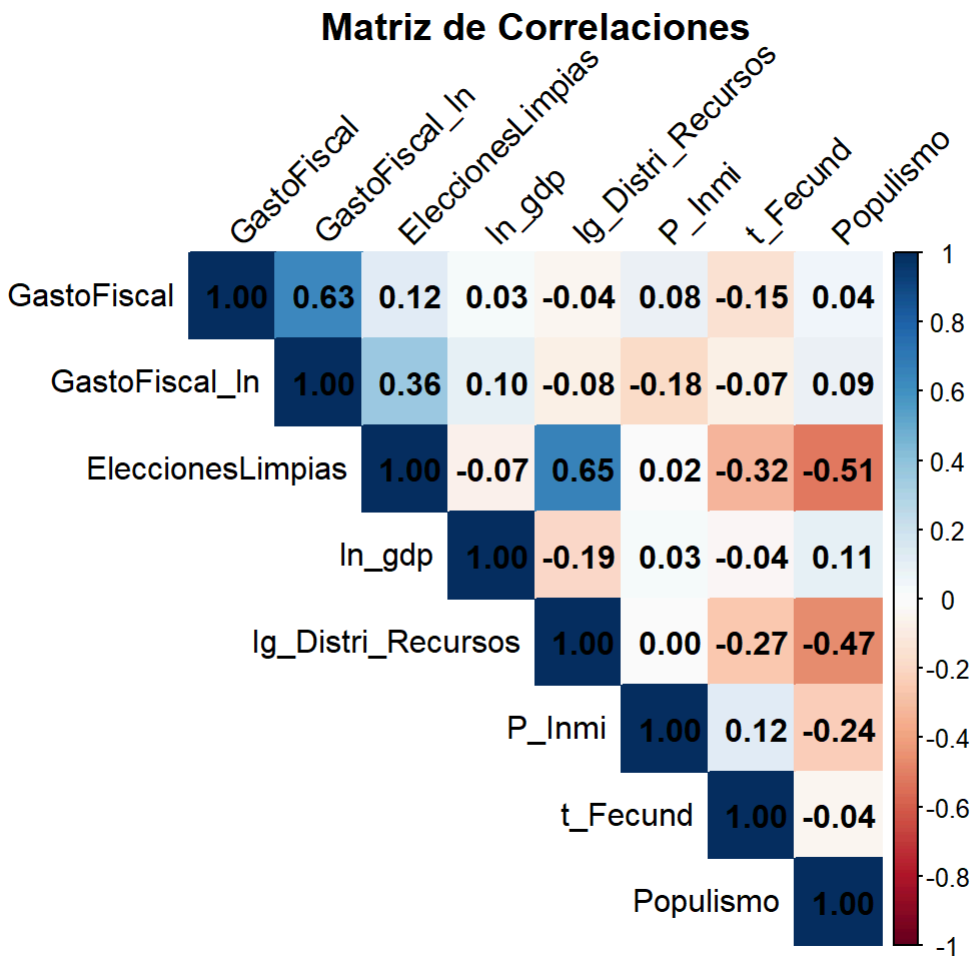


1.3 Matriz de correlaciones

```
df.num <- df[, c(GastoFiscal, GastoFiscal_ln, EleccionesLimpias, ln_gdp,
                 Ig_Distri_Recursos, P_Inmi, t_Fecund, Populismo)]
cor_matrix <- cor(df.num, use = "complete.obs")
round(cor_matrix, 3)
```

```
##                    GastoFiscal GastoFiscal_ln EleccionesLimpias ln_gdp
## GastoFiscal              1.000          0.631             0.118  0.026
## GastoFiscal_ln           0.631          1.000             0.358  0.095
## EleccionesLimpias        0.118          0.358             1.000 -0.067
## ln_gdp                   0.026          0.095            -0.067  1.000
## Ig_Distri_Recursos      -0.043         -0.078             0.651 -0.191
## P_Inmi                   0.081         -0.184             0.018  0.025
## t_Fecund                -0.146         -0.074            -0.320 -0.039
## Populismo                0.041          0.087            -0.510  0.109
##                    Ig_Distri_Recursos P_Inmi t_Fecund Populismo
## GastoFiscal                    -0.043  0.081   -0.146     0.041
## GastoFiscal_ln                 -0.078 -0.184   -0.074     0.087
## EleccionesLimpias               0.651  0.018   -0.320    -0.510
## ln_gdp                         -0.191  0.025   -0.039     0.109
## Ig_Distri_Recursos              1.000 -0.001   -0.270    -0.469
## P_Inmi                         -0.001  1.000    0.118    -0.238
## t_Fecund                       -0.270  0.118    1.000    -0.043
## Populismo                      -0.469 -0.238   -0.043     1.000
```

```
corrplot(cor_matrix, method = "color", type = "upper",
         addCoef.col = "black", tl.col = "black", tl.srt = 45,
         title = "Matriz de Correlaciones", mar = c(0,0,1,0))
```



2. Estimación de los modelos MCO

```
# MODELO 1: Lineal
# Y = β0 + β1·frefair + β2·ln_gdp + β3·eqdr + β4·immig + u


formula1 <- as.formula(paste(GastoFiscal, "~", EleccionesLimpias, "+", ln_gdp, "+", Ig_Distri
_Recursos, "+", P_Inmi))
mod1 <- lm(formula1, data = df)
summary(mod1)
```

```
##
## Call:
## lm(formula = formula1, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5998 -0.4219 -0.1402  0.0024  3.7775
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -0.273898   2.924570  -0.094    0.926
## EleccionesLimpias   1.814234   1.738121   1.044    0.305
## ln_gdp              0.001068   0.096856   0.011    0.991
## Ig_Distri_Recursos -1.008687   1.197827  -0.842    0.407
## P_Inmi              0.031519   0.075236   0.419    0.678
##
## Residual standard error: 0.8513 on 29 degrees of freedom
## Multiple R-squared:  0.04454,    Adjusted R-squared:  -0.08725
## F-statistic: 0.3379 on 4 and 29 DF,  p-value: 0.8501
```

```
# MODELO GANADOR: Log-Lineal (M2)
# ln(Y) = β0 + β1·frefair + β2·ln_gdp + β3·eqdr + β4·immig + u



formula2 <- as.formula(paste(GastoFiscal_ln, "~", EleccionesLimpias, "+", ln_gdp, "+", Ig_Dis
tri_Recursos, "+", P_Inmi))
mod2 <- lm(formula2, data = df)
summary(mod2)
```

```
##
## Call:
## lm(formula = formula2, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.7009 -1.1921  0.1998  0.9801  2.9187
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -8.30366    5.24481  -1.583  0.12422
## EleccionesLimpias  11.12042    3.11708   3.568  0.00128 **
## ln_gdp              0.05194    0.17370   0.299  0.76704
## Ig_Distri_Recursos -5.64705    2.14813  -2.629  0.01356 *
## P_Inmi             -0.17724    0.13493  -1.314  0.19928
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.527 on 29 degrees of freedom
## Multiple R-squared:  0.338,  Adjusted R-squared:  0.2467
## F-statistic: 3.702 on 4 and 29 DF,  p-value: 0.0149
```

```
# MODELO 3: Polinómico (cuadrático en democracia) — Log-lineal
# ln(Y) = β0 + β1·frefair + β2·frefair² + β3·ln_gdp + β4·miferrat + β5·popul + u
# Permite captar una relación en forma de U o U invertida


# Creamos el término cuadrático de la variable de interés
df$interes2 <- df[[EleccionesLimpias]]^2

formula3 <- as.formula(paste(GastoFiscal_ln, "~", EleccionesLimpias, "+ interes2 +", ln_gdp,
"+", Ig_Distri_Recursos, "+", P_Inmi))
mod3 <- lm(formula3, data = df)
summary(mod3)
```

```
##
## Call:
## lm(formula = formula3, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.5953 -0.8756  0.2018  0.8582  3.1612
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -20.9975    11.0334  -1.903   0.0674 .
## EleccionesLimpias   43.4813    25.0219   1.738   0.0932 .
## interes2           -23.0902    17.7178  -1.303   0.2031
## ln_gdp               0.0895     0.1741   0.514   0.6111
## Ig_Distri_Recursos  -3.6927     2.5990  -1.421   0.1664
## P_Inmi              -0.1367     0.1369  -0.998   0.3268
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.509 on 28 degrees of freedom
## Multiple R-squared:  0.3759, Adjusted R-squared:  0.2644
## F-statistic: 3.373 on 5 and 28 DF,  p-value: 0.01649
```

3. Diagnóstico y chequeo de los modelos

```
# Guardamos los residuos de cada modelo
res1 <- resid(mod1)
res2 <- resid(mod2)
res3 <- resid(mod3)

yhat1 <- fitted(mod1)
yhat2 <- fitted(mod2)
yhat3 <- fitted(mod3)
```

3.1 Estadísticos de los residuos

```
# Función auxiliar para resumir residuos
resumen_res <- function(res, nombre) {
  cat("===", nombre, "===\n")
  cat(round(c(Media     = mean(res),
              Mediana   = median(res),
              Máximo    = max(res),
              Mínimo    = min(res),
              Desv.Tip  = sd(res),
              Asimetría = skewness(res),
              Curtosis  = kurtosis(res)), 4), "\n\n")
}

resumen_res(res1, "Modelo 1 — Lineal")
```

```
## === Modelo 1 — Lineal ===
## 0 -0.1402 3.7775 -0.5998 0.798 3.3218 12.3805
```

```
resumen_res(res2, "Modelo 2 — Log-lineal")
```

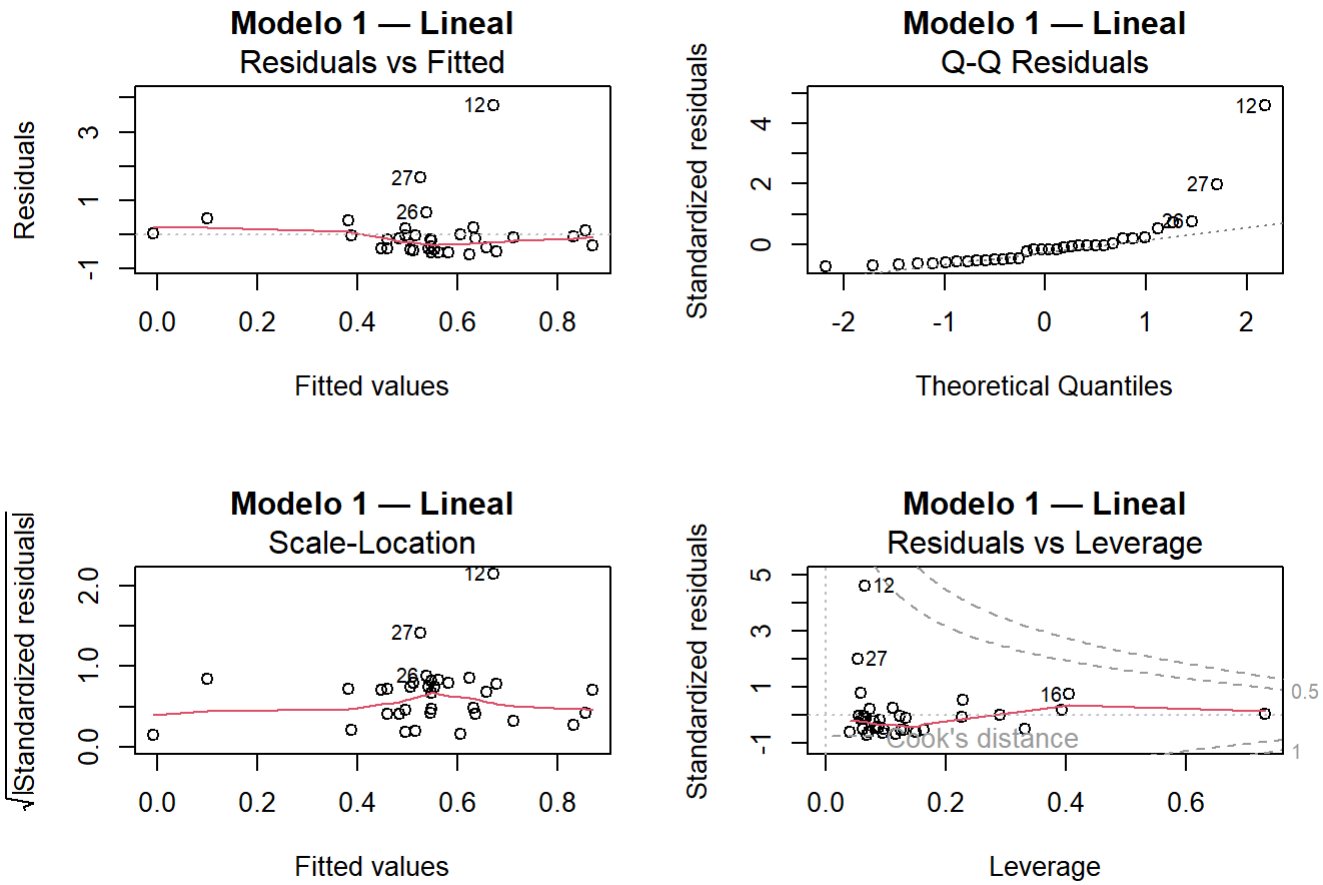
```
## === Modelo 2 — Log-lineal ===
## 0 0.1998 2.9187 -2.7009 1.4311 -0.031 -0.8089
```

```
resumen_res(res3, "Modelo 3 — Cuadrático")
```

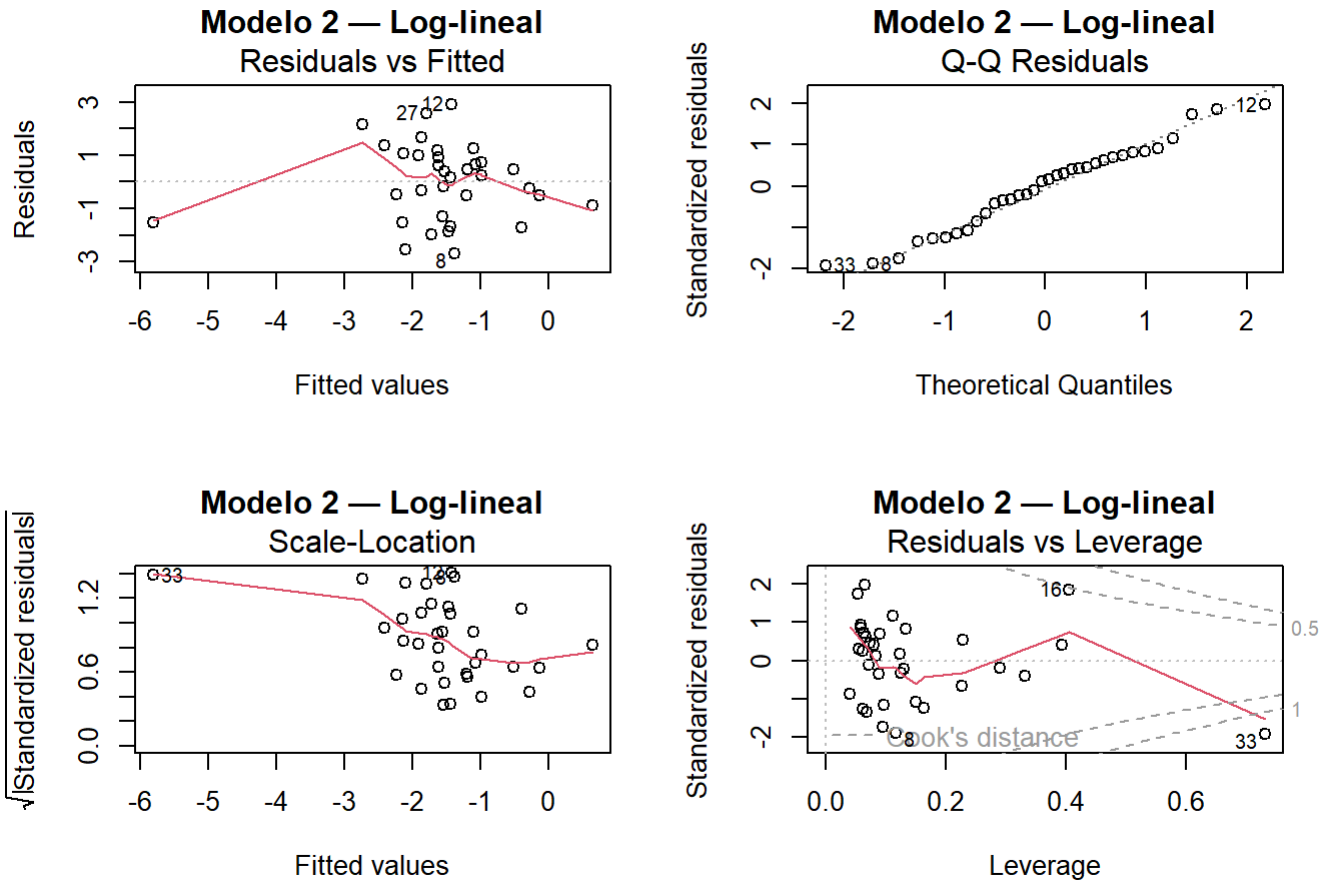
```
## === Modelo 3 — Cuadrático ===
## 0 0.2018 3.1612 -2.5953 1.3896 0.0082 -0.5213
```

3.2 Gráficos de residuos

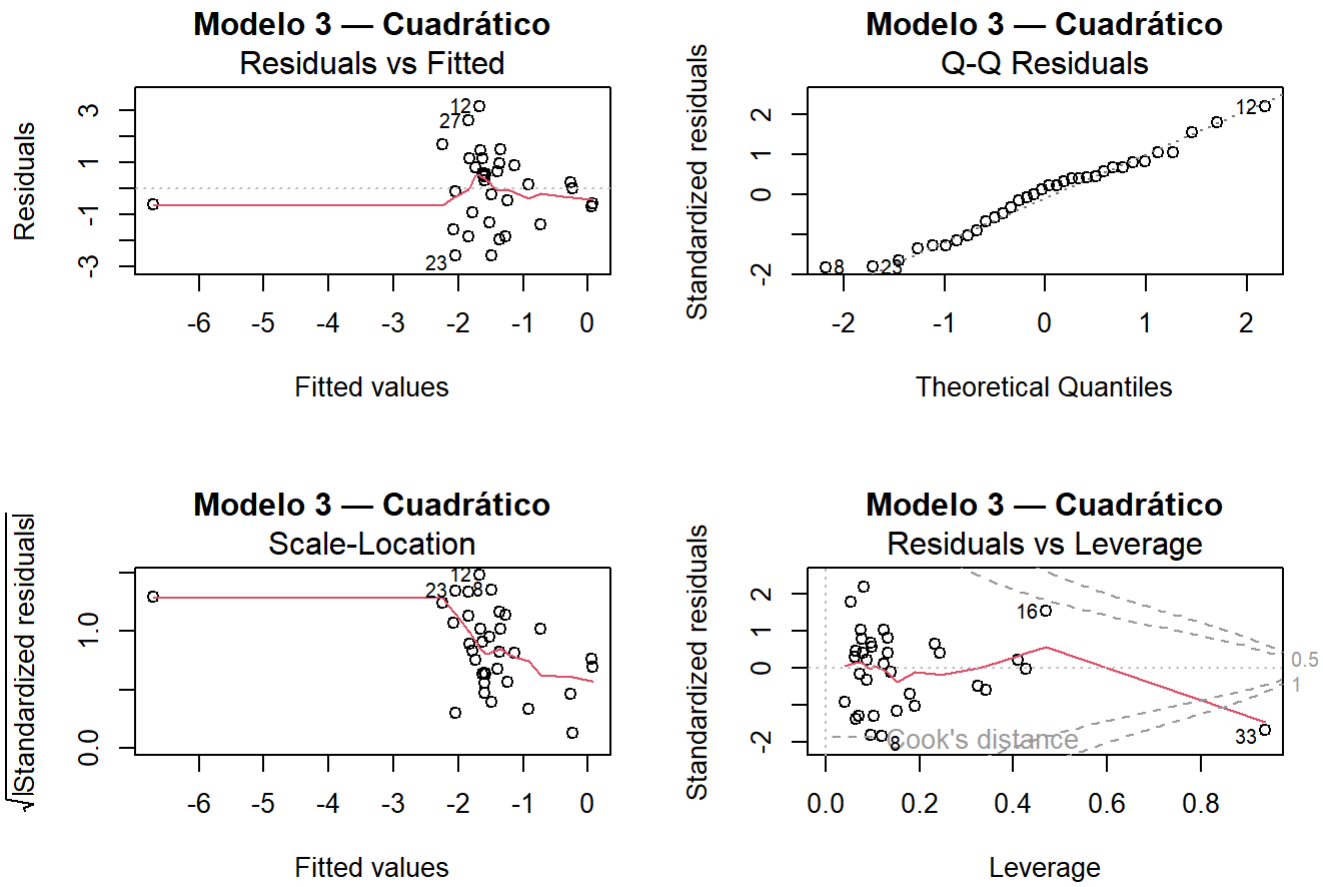
```
par(mfrow = c(2, 2))
plot(mod1, main = "Modelo 1 — Lineal")
```



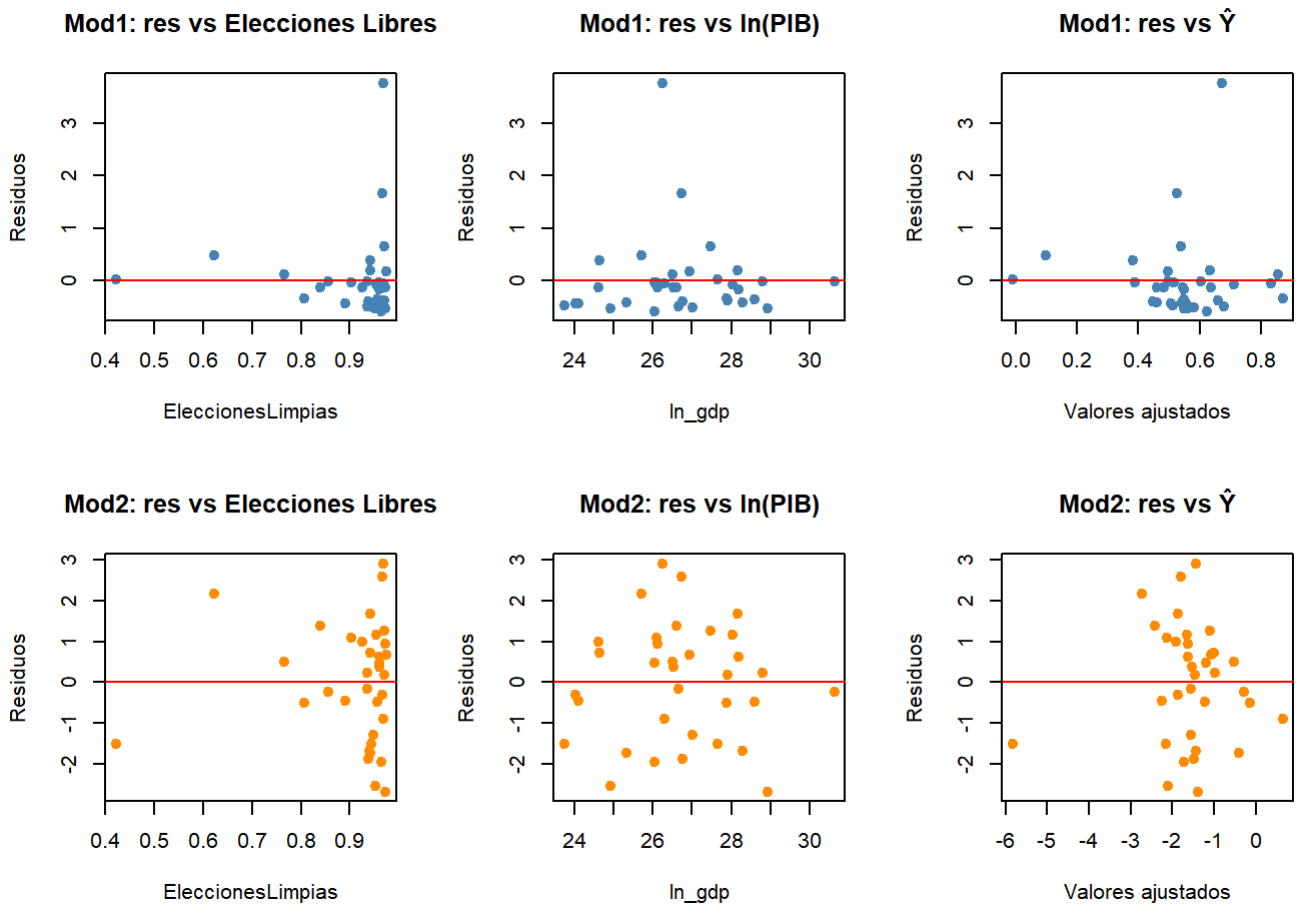

```
par(mfrow = c(2, 2))
plot(mod2, main = "Modelo 2 — Log-lineal")
```




```
par(mfrow = c(2, 2))
plot(mod3, main = "Modelo 3 — Cuadrático")
```



```
# Ventana gráfica de 2 filas y 3 columnas
par(mfrow = c(2, 3))
# -- MODELO 1 --
plot(x = df$EleccionesLimpias, y = res1, main = "Mod1: res vs Elecciones Libres",
     xlab = "EleccionesLimpias", ylab = "Residuos", pch = 19, col = "steelblue")
abline(h = 0, col = "red")

plot(x = df$ln_gdp, y = res1, main = "Mod1: res vs ln(PIB)",
     xlab = "ln_gdp", ylab = "Residuos", pch = 19, col = "steelblue")
abline(h = 0, col = "red")

plot(x = yhat1, y = res1, main = "Mod1: res vs Ŷ",
     xlab = "Valores ajustados", ylab = "Residuos", pch = 19, col = "steelblue")
abline(h = 0, col = "red")

# -- MODELO 2 --
plot(x = df$EleccionesLimpias, y = res2, main = "Mod2: res vs Elecciones Libres",
     xlab = "EleccionesLimpias", ylab = "Residuos", pch = 19, col = "darkorange")
abline(h = 0, col = "red")

plot(x = df$ln_gdp, y = res2, main = "Mod2: res vs ln(PIB)",
     xlab = "ln_gdp", ylab = "Residuos", pch = 19, col = "darkorange")
abline(h = 0, col = "red")

plot(x = yhat2, y = res2, main = "Mod2: res vs Ŷ",
     xlab = "Valores ajustados", ylab = "Residuos", pch = 19, col = "darkorange")
abline(h = 0, col = "red")
```



3.3 Contraste de Heterocedasticidad

Breusch-Pagan

```
cat("=== Contraste de Breusch-Pagan ===\n")
```

```
## === Contraste de Breusch-Pagan ===
```

```
cat("H0: Homocedasticidad  |  H1: Heterocedasticidad\n\n")
```

```
## H0: Homocedasticidad  |  H1: Heterocedasticidad
```

```
bp1 <- bptest(mod1)
bp2 <- bptest(mod2)
bp3 <- bptest(mod3)

cat("Modelo 1 — p-valor:", round(bp1$p.value, 4),
    ifelse(bp1$p.value < 0.05, "→ RECHAZA H0 (heterocedasticidad)", "→ No rechaza H0"), "\n")
```

```
## Modelo 1 — p-valor: 0.8284 → No rechaza H0
```

```
cat("Modelo 2 — p-valor:", round(bp2$p.value, 4),
    ifelse(bp2$p.value < 0.05, "→ RECHAZA H0 (heterocedasticidad)", "→ No rechaza H0"), "\n")
```

```
## Modelo 2 — p-valor: 0.6475 → No rechaza H0
```

```
cat("Modelo 3 — p-valor:", round(bp3$p.value, 4),
    ifelse(bp3$p.value < 0.05, "→ RECHAZA H0 (heterocedasticidad)", "→ No rechaza H0"), "\n")
```

```
## Modelo 3 — p-valor: 0.6572 → No rechaza H0
```

White

```
cat("=== Contraste de White ===\n")
```

```
## === Contraste de White ===
```

```
cat("(versión no constructiva — incluye cuadrados y productos cruzados)\n")
```

```
## (versión no constructiva — incluye cuadrados y productos cruzados)
```

```
cat("H0: Homocedasticidad  |  H1: Heterocedasticidad\n\n")
```

```
## H0: Homocedasticidad  |  H1: Heterocedasticidad
```

```
white1 <- bptest(mod1, ~ EleccionesLimpias * ln_gdp + I(EleccionesLimpias^2) + I(ln_gdp^2), d
ata = df)
white2 <- bptest(mod2, ~ EleccionesLimpias * ln_gdp + I(EleccionesLimpias^2) + I(ln_gdp^2), d
ata = df)
white3 <- bptest(mod3, ~ EleccionesLimpias * ln_gdp + I(EleccionesLimpias^2) + I(ln_gdp^2), d
ata = df)

cat("Modelo 1 — p-valor:", round(white1$p.value, 4),
    ifelse(white1$p.value < 0.05, "→ RECHAZA H0", "→ No rechaza H0"), "\n")
```

```
## Modelo 1 — p-valor: 0.8945 → No rechaza H0
```

```
cat("Modelo 2 — p-valor:", round(white2$p.value, 4),
    ifelse(white2$p.value < 0.05, "→ RECHAZA H0", "→ No rechaza H0"), "\n")
```

```
## Modelo 2 — p-valor: 0.6059 → No rechaza H0
```

```
cat("Modelo 3 — p-valor:", round(white3$p.value, 4),
    ifelse(white3$p.value < 0.05, "→ RECHAZA H0", "→ No rechaza H0"), "\n")
```

```
## Modelo 3 — p-valor: 0.6294 → No rechaza H0
```

Goldfeld-Quandt

```
cat("=== Contraste de Goldfeld-Quandt ===\n")
```

```
## === Contraste de Goldfeld-Quandt ===
```

```
cat("(ordenado por ln_gdp — variable de control)\n")
```

```
## (ordenado por ln_gdp — variable de control)
```

```
cat("H0: Homocedasticidad  |  H1: Heterocedasticidad\n\n")
```

```
## H0: Homocedasticidad  |  H1: Heterocedasticidad
```

```
gq1 <- gqtest(mod1, order.by = ~ ln_gdp, data = df, fraction = 0.2)
gq2 <- gqtest(mod2, order.by = ~ ln_gdp, data = df, fraction = 0.2)
gq3 <- gqtest(mod3, order.by = ~ ln_gdp, data = df, fraction = 0.2)

cat("Modelo 1 — p-valor:", round(gq1$p.value, 4),
    ifelse(gq1$p.value < 0.05, "→ RECHAZA H0", "→ No rechaza H0"), "\n")
```

```
## Modelo 1 — p-valor: 0.9991 → No rechaza H0
```

```
cat("Modelo 2 — p-valor:", round(gq2$p.value, 4),
    ifelse(gq2$p.value < 0.05, "→ RECHAZA H0", "→ No rechaza H0"), "\n")
```

```
## Modelo 2 — p-valor: 0.7349 → No rechaza H0
```

```
cat("Modelo 3 — p-valor:", round(gq3$p.value, 4),
    ifelse(gq3$p.value < 0.05, "→ RECHAZA H0", "→ No rechaza H0"), "\n")
```

```
## Modelo 3 — p-valor: 0.7296 → No rechaza H0
```

Estimación robusta (corrección si hay heterocedasticidad)

```
cat("=== Estimación con errores estándar robustos HC2 ===\n\n")
```

```
## === Estimación con errores estándar robustos HC2 ===
```

```
cat(" Modelo 1 \n")
```

```
##  Modelo 1
```

```
coeftest(mod1, vcov. = vcovHC, type = "HC2")
```

```
##
## t test of coefficients:
##
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -0.2738984  1.4404956 -0.1901  0.85052
## EleccionesLimpias   1.8142338  0.8920121  2.0339  0.05120 .
## ln_gdp              0.0010678  0.0506805  0.0211  0.98333
## Ig_Distri_Recursos -1.0086874  0.5238404 -1.9256  0.06402 .
## P_Inmi              0.0315192  0.0647737  0.4866  0.63020
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
cat("\n Modelo 2 \n")
```

```
##
##  Modelo 2
```

```
coeftest(mod2, vcov. = vcovHC, type = "HC2")
```

```
##
## t test of coefficients:
##
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -8.303657   5.625991 -1.4759 0.150735
## EleccionesLimpias  11.120421   5.213421  2.1330 0.041511 *
## ln_gdp              0.051943   0.158321  0.3281 0.745205
## Ig_Distri_Recursos -5.647053   1.831818 -3.0828 0.004468 **
## P_Inmi             -0.177237   0.154894 -1.1443 0.261876
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
cat("\n Modelo 3 \n")
```

```
##
##  Modelo 3
```

```
coeftest(mod3, vcov. = vcovHC, type = "HC2")
```

```
##
## t test of coefficients:
##
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -20.997467  11.362541 -1.8480  0.07520 .
## EleccionesLimpias   43.481331  26.955306  1.6131  0.11794
## interes2           -23.090199  17.625795 -1.3100  0.20083
## ln_gdp               0.089504   0.155454  0.5758  0.56938
## Ig_Distri_Recursos  -3.692661   1.658031 -2.2271  0.03415 *
## P_Inmi              -0.136661   0.129245 -1.0574  0.29938
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3.4 Contraste de Normalidad — Jarque-Bera

```
cat("=== Contraste de Jarque-Bera ===\n")
```

```
## === Contraste de Jarque-Bera ===
```

```
cat("H0: Residuos normales  |  H1: No normalidad\n\n")
```

```
## H0: Residuos normales  |  H1: No normalidad
```

```
jb1 <- jarque.bera.test(res1)
jb2 <- jarque.bera.test(res2)
jb3 <- jarque.bera.test(res3)

cat("Modelo 1 — p-valor:", round(jb1$p.value, 4),
    ifelse(jb1$p.value < 0.05, "→ RECHAZA H0 (no normalidad)", "→ No rechaza H0 (normalida
d)"), "\n")
```

```
## Modelo 1 — p-valor: 0 → RECHAZA H0 (no normalidad)
```

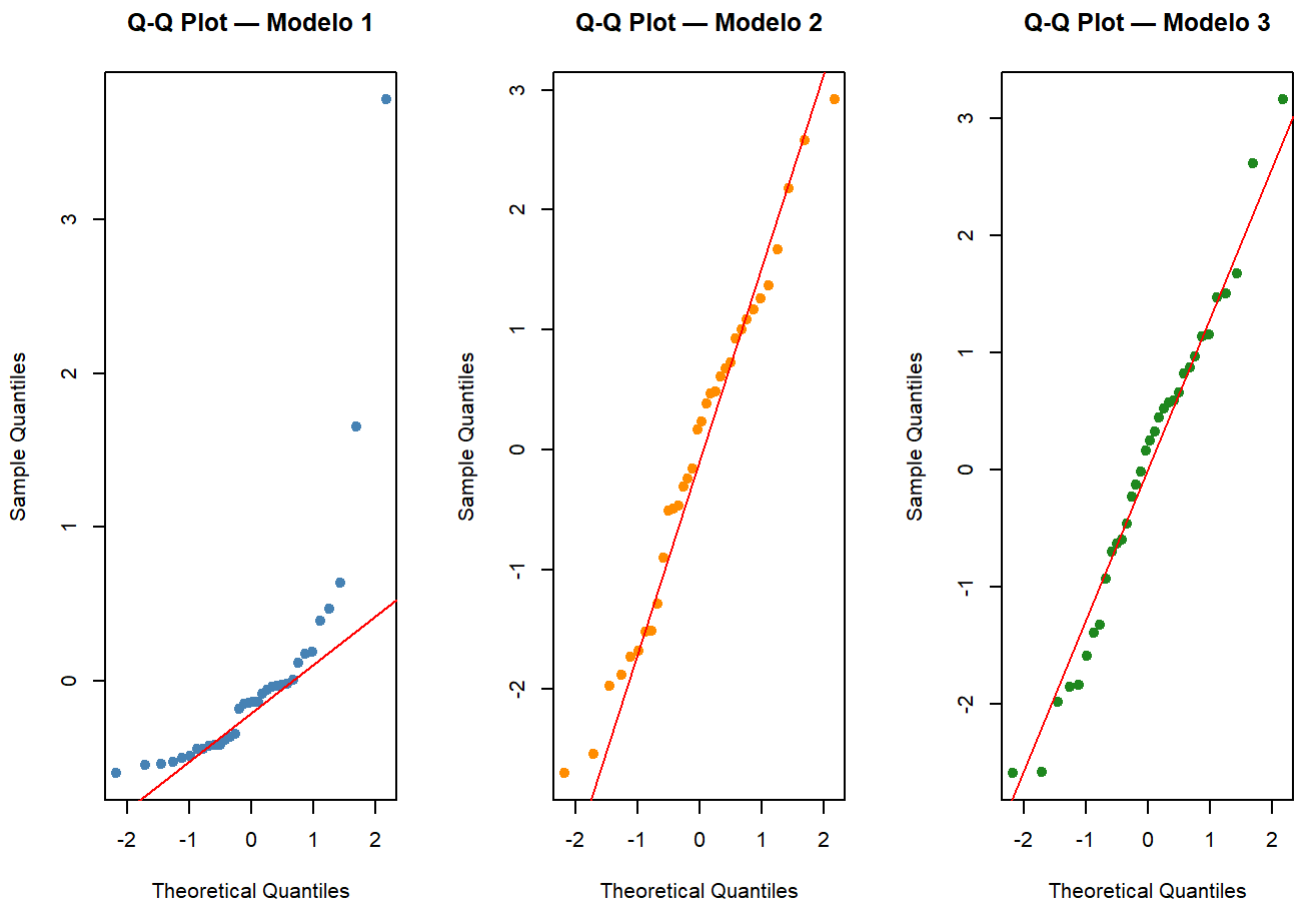
```
cat("Modelo 2 — p-valor:", round(jb2$p.value, 4),
    ifelse(jb2$p.value < 0.05, "→ RECHAZA H0 (no normalidad)", "→ No rechaza H0 (normalida
d)"), "\n")
```

```
## Modelo 2 — p-valor: 0.7226 → No rechaza H0 (normalidad)
```

```
cat("Modelo 3 — p-valor:", round(jb3$p.value, 4),
    ifelse(jb3$p.value < 0.05, "→ RECHAZA H0 (no normalidad)", "→ No rechaza H0 (normalida
d)"), "\n")
```

```
## Modelo 3 — p-valor: 0.908 → No rechaza H0 (normalidad)
```

```
par(mfrow = c(1, 3))

qqnorm(res1, main = "Q-Q Plot — Modelo 1", pch = 19, col = "steelblue")
qqline(res1, col = "red")

qqnorm(res2, main = "Q-Q Plot — Modelo 2", pch = 19, col = "darkorange")
qqline(res2, col = "red")

qqnorm(res3, main = "Q-Q Plot — Modelo 3", pch = 19, col = "forestgreen")
qqline(res3, col = "red")
```



3.5 Contraste de Forma Funcional — REsET de Ramsey

```
cat("=== Contraste REsET de Ramsey ===\n")
```

```
## === Contraste REsET de Ramsey ===
```

```
cat("H0: Forma funcional correcta  |  H1: Especificación incorrecta\n\n")
```

```
## H0: Forma funcional correcta  |  H1: Especificación incorrecta
```

```
reset1 <- resettest(mod1, power = 2:3, type = "fitted")
reset2 <- resettest(mod2, power = 2:3, type = "fitted")
reset3 <- resettest(mod3, power = 2:3, type = "fitted")

cat("Modelo 1 — p-valor:", round(reset1$p.value, 4),
    ifelse(reset1$p.value < 0.05, "→ RECHAZA H0 (mala especificación)", "→ No rechaza H0"),
"\n")
```

```
## Modelo 1 — p-valor: 0.597 → No rechaza H0
```

```
cat("Modelo 2 — p-valor:", round(reset2$p.value, 4),
    ifelse(reset2$p.value < 0.05, "→ RECHAZA H0 (mala especificación)", "→ No rechaza H0"),
"\n")
```

```
## Modelo 2 — p-valor: 0.1514 → No rechaza H0
```

```
cat("Modelo 3 — p-valor:", round(reset3$p.value, 4),
    ifelse(reset3$p.value < 0.05, "→ RECHAZA H0 (mala especificación)", "→ No rechaza H0"),
"\n")
```

```
## Modelo 3 — p-valor: 0.1602 → No rechaza H0
```

3.6 Multicolinealidad — Factor de Inflación de la Varianza (VIF)

```
cat("=== VIF — Factor de Inflación de la Varianza ===\n")
```

```
## === VIF — Factor de Inflación de la Varianza ===
```

```
cat("Regla práctica: VIF > 10 indica multicolinealidad severa\n\n")
```

```
## Regla práctica: VIF > 10 indica multicolinealidad severa
```

```
cat("Modelo 1:\n")
```

```
## Modelo 1:
```

```
vif(mod1)
```

```
##  EleccionesLimpias             ln_gdp Ig_Distri_Recursos             P_Inmi
##           1.747021           1.044712           1.804433           1.001176
```

```
cat("\nModelo 2:\n")
```

```
##
## Modelo 2:
```

```
vif(mod2)
```

```
##  EleccionesLimpias             ln_gdp Ig_Distri_Recursos             P_Inmi
##           1.747021           1.044712           1.804433           1.001176
```

```
cat("\nModelo 3 (ojo: dem y dem² serán colineales — es estructural):\n")
```

```
##
## Modelo 3 (ojo: dem y dem² serán colineales — es estructural):
```

```
vif(mod3)
```

```
##  EleccionesLimpias           interes2             ln_gdp Ig_Distri_Recursos
##         115.286206         128.476907           1.074168           2.705054
##             P_Inmi
##           1.055774
```

```
# Nota: En el Mod3 la colinealidad entre dem y dem² es esperada y
# estructural del modelo cuadrático. No invalida el modelo.
```

3.7 Observaciones Influyentes — Distancia de Cook y Leverage

```
# Tabla de observaciones potencialmente influyentes
cat("=== Observaciones con Cook > 4/n ===\n")
```

```
## === Observaciones con Cook > 4/n ===
```

```
cat("Umbral Cook:", round(4/nrow(df), 4), "\n\n")
```

```
## Umbral Cook: 0.1176
```

```
influyentes <- df %>%
  mutate(cook1     = cooks.distance(mod1),
         cook2     = cooks.distance(mod2),
         leverage1 = hatvalues(mod1),
         leverage2 = hatvalues(mod2)) %>%
  filter(cook1 > 4/nrow(df) | cook2 > 4/nrow(df)) %>%
  select(iso3_gted, GastoFiscal_ln, EleccionesLimpias, ln_gdp,
         cook1, cook2, leverage1, leverage2)

print(influyentes)
```

```
## # A tibble: 3 × 8
##   iso3_gted GastoFiscal_ln EleccionesLimpias ln_gdp    cook1  cook2 leverage1
##   <chr>              <dbl>             <dbl>  <dbl>    <dbl>  <dbl>     <dbl>
## 1 FIN                1.49              0.969   26.3 0.295    0.0547    0.0654
## 2 HUN               -0.559             0.622   25.7 0.0708   0.468     0.406
## 3 TUR               -7.34              0.422   27.6 0.000229 2.02      0.732
## # ℹ 1 more variable: leverage2 <dbl>
```

```
# Robustez del Modelo 2: exclusión de Turquía vs. dummy de Turquía


#  Versión 1: excluir Turquía
df_sin_tur <- df[df$iso3_gted != "TUR", ]

mod2_sin_tur <- lm(as.formula(paste(GastoFiscal_ln, "~", EleccionesLimpias, "+",
                                    ln_gdp, "+", Ig_Distri_Recursos, "+", P_Inmi)),
                   data = df_sin_tur)

#  Versión 2: dummy de Turquía (muestra completa)
df$d_tur <- as.integer(df$iso3_gted == "TUR")

mod2_dummy_tur <- lm(as.formula(paste(GastoFiscal_ln, "~", EleccionesLimpias, "+",
                                      ln_gdp, "+", Ig_Distri_Recursos, "+", P_Inmi,
                                      "+ d_tur")),
                     data = df)

#  Resultados individuales
cat("=== Modelo 2 — Sin Turquía ===\n")
```

```
## === Modelo 2 — Sin Turquía ===
```

```
summary(mod2_sin_tur)
```

```
##
## Call:
## lm(formula = as.formula(paste(GastoFiscal_ln, "~", EleccionesLimpias,
##     "+", ln_gdp, "+", Ig_Distri_Recursos, "+", P_Inmi)), data = df_sin_tur)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.7422 -0.8007  0.1885  0.9091  3.0446
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -3.89141    5.44114  -0.715    0.480
## EleccionesLimpias   2.22727    5.29941   0.420    0.677
## ln_gdp              0.12016    0.16851   0.713    0.482
## Ig_Distri_Recursos -3.23492    2.36421  -1.368    0.182
## P_Inmi             -0.07609    0.13764  -0.553    0.585
##
## Residual standard error: 1.451 on 28 degrees of freedom
## Multiple R-squared:  0.1274, Adjusted R-squared:  0.002792
## F-statistic: 1.022 on 4 and 28 DF,  p-value: 0.4129
```

```
cat("=== Modelo 2 — Dummy Turquía ===\n")
```

```
## === Modelo 2 — Dummy Turquía ===
```

```
summary(mod2_dummy_tur)
```

```
##
## Call:
## lm(formula = as.formula(paste(GastoFiscal_ln, "~", EleccionesLimpias,
##     "+", ln_gdp, "+", Ig_Distri_Recursos, "+", P_Inmi, "+ d_tur")),
##     data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.7422 -0.7583  0.1671  0.8705  3.0446
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -3.89141    5.44114  -0.715   0.4804
## EleccionesLimpias   2.22727    5.29941   0.420   0.6775
## ln_gdp              0.12016    0.16851   0.713   0.4817
## Ig_Distri_Recursos -3.23492    2.36421  -1.368   0.1821
## P_Inmi             -0.07609    0.13764  -0.553   0.5848
## d_tur              -5.67136    2.80199  -2.024   0.0526 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.451 on 28 degrees of freedom
## Multiple R-squared:  0.4225, Adjusted R-squared:  0.3194
## F-statistic: 4.097 on 5 and 28 DF,  p-value: 0.006444
```

```
#  Tabla comparativa de coeficientes
cat("\n=== Comparación de coeficientes: Modelo 2 base vs. robustez ===\n\n")
```

```
##
## === Comparación de coeficientes: Modelo 2 base vs. robustez ===
```

```
cat(sprintf("%-25s %-14s %-14s %-14s\n", "Variable", "Base", "Sin TUR", "Dummy TUR"))
```

```
## Variable                  Base           Sin TUR        Dummy TUR
```

```
cat(rep("-", 68), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
- - - - - - - - - - - - - - - - - - - - - - - - -
```

```
vars_base <- names(coef(mod2))
for (v in vars_base) {
  b_base    <- coef(mod2)[v]
  b_sintur  <- ifelse(v %in% names(coef(mod2_sin_tur)),
                      coef(mod2_sin_tur)[v], NA)
  b_dummy   <- ifelse(v %in% names(coef(mod2_dummy_tur)),
                      coef(mod2_dummy_tur)[v], NA)
  cat(sprintf("%-25s %-14.4f %-14s %-14s\n", v,
              b_base,
              ifelse(is.na(b_sintur), "  —", sprintf("%.4f", b_sintur)),
              ifelse(is.na(b_dummy),  "  —", sprintf("%.4f", b_dummy))))
}
```

```
## (Intercept)               -8.3037        -3.8914        -3.8914
## EleccionesLimpias         11.1204        2.2273         2.2273
## ln_gdp                    0.0519         0.1202         0.1202
## Ig_Distri_Recursos        -5.6471        -3.2349        -3.2349
## P_Inmi                    -0.1772        -0.0761        -0.0761
```

```
# Coeficiente de la dummy (no está en el modelo base)
cat(sprintf("%-25s %-14s %-14s %-14.4f\n", "d_tur",
            "  —", "  —", coef(mod2_dummy_tur)["d_tur"]))
```

```
## d_tur                       —           —          -5.6714
```

4. selección del modelo

```
# Comparamos R² ajustado y AIC/BIC como criterios adicionales
cat("=== Criterios de selección de modelo ===\n\n")
```

```
## === Criterios de selección de modelo ===
```

```
cat(sprintf("%-25s %-12s %-12s %-12s\n", "Criterio", "Modelo 1", "Modelo 2", "Modelo 3"))
```

```
## Criterio                  Modelo 1     Modelo 2     Modelo 3
```

```
cat(sprintf("%-25s %-12.4f %-12.4f %-12.4f\n", "R² ajustado",
            summary(mod1)$adj.r.squared,
            summary(mod2)$adj.r.squared,
            summary(mod3)$adj.r.squared))
```

```
## R² ajustado              -0.0873      0.2467       0.2644
```

```
cat(sprintf("%-25s %-12.2f %-12.2f %-12.2f\n", "AIC",
            AIC(mod1), AIC(mod2), AIC(mod3)))
```

```
## AIC                       92.13        131.85       131.85
```

```
cat(sprintf("%-25s %-12.2f %-12.2f %-12.2f\n", "BIC",
            BIC(mod1), BIC(mod2), BIC(mod3)))
```

```
## BIC                       101.29       141.01       142.53
```

```
cat(sprintf("%-25s %-12.4f %-12.4f %-12.4f\n", "p-val REsET",
            reset1$p.value, reset2$p.value, reset3$p.value))
```

```
## p-val REsET               0.5970       0.1514       0.1602
```

```
cat(sprintf("%-25s %-12.4f %-12.4f %-12.4f\n", "p-val Breusch-Pagan",
            bp1$p.value, bp2$p.value, bp3$p.value))
```

```
## p-val Breusch-Pagan       0.8284       0.6475       0.6572
```

```
cat(sprintf("%-25s %-12.4f %-12.4f %-12.4f\n", "p-val Jarque-Bera",
            jb1$p.value, jb2$p.value, jb3$p.value))
```

```
## p-val Jarque-Bera         0.0000       0.7226       0.9080
```

5. Predicción con el modelo seleccionado

```
# Predicción con Modelo 2 (log-lineal)
# ln(Y) = β0 + β1·frefair + β2·ln_gdp + β3·eqdr + β4·immig + u


# País hipotético — ajusta los valores según el escenario deseado
nuevo_pais <- data.frame(
  EleccionesLimpias = 0.75,
  ln_gdp        = log(30000),
  Ig_Distri_Recursos    = mean(df$Ig_Distri_Recursos, na.rm = TRUE),   # 0.848
  P_Inmi     = mean(df$P_Inmi,  na.rm = TRUE)    # -0.183
)

cat("=== Predicción puntual — Modelo 2 (log-lineal) ===\n")
```

```
## === Predicción puntual — Modelo 2 (log-lineal) ===
```

```
cat("País hipotético: EleccionesLimpias = 0.75 | ln_gdp =", round(log(30000), 3),
    "| Ig_Distri_Recursos = 0 | P_Inmi = 0\n\n")
```

```
## País hipotético: EleccionesLimpias = 0.75 | ln_gdp = 10.309 | Ig_Distri_Recursos = 0 | P_I
nmi = 0
```

```
pred_media <- predict(mod2, newdata = nuevo_pais, interval = "confidence", level = 0.95)
pred_indiv <- predict(mod2, newdata = nuevo_pais, interval = "prediction", level = 0.95)

cat("Intervalo de confianza (media):\n")
```

```
## Intervalo de confianza (media):
```

```
cat("  ln(TE_CIT): [", round(pred_media[2], 4), ",", round(pred_media[3], 4), "]\n")
```

```
##   ln(TE_CIT): [ -10.048 , 1.6781 ]
```

```
cat("  TE_CIT (anti-log): [", round(exp(pred_media[2]), 4), ",", round(exp(pred_media[3]),
4), "] % del PIB\n\n")
```

```
##   TE_CIT (anti-log): [ 0 , 5.3552 ] % del PIB
```

```
cat("Intervalo de predicción (observación individual):\n")
```

```
## Intervalo de predicción (observación individual):
```

```
cat("  ln(TE_CIT): [", round(pred_indiv[2], 4), ",", round(pred_indiv[3], 4), "]\n")
```

```
##   ln(TE_CIT): [ -10.8276 , 2.4576 ]
```

```
cat("  TE_CIT (anti-log): [", round(exp(pred_indiv[2]), 4), ",", round(exp(pred_indiv[3]),
4), "] % del PIB\n")
```

```
##   TE_CIT (anti-log): [ 0 , 11.6768 ] % del PIB
```

6. Interpretación de coeficientes

```
cat("=== Interpretación de los coeficientes ===\n\n")
```

```
## === Interpretación de los coeficientes ===
```

```
# Modelo 1
b1_m1 <- coef(mod1)["EleccionesLimpias"]
cat("Modelo 1 (lineal):\n")
```

```
## Modelo 1 (lineal):
```

```
cat("  β₁ =", round(b1_m1, 4), "\n")
```

```
##   β₁ = 1.8142
```

```
cat("  Interpretación: Un aumento de 1 punto en EleccionesLimpias se asocia con\n")
```

```
##   Interpretación: Un aumento de 1 punto en EleccionesLimpias se asocia con
```

```
cat("  un cambio de", round(b1_m1, 4), "puntos porcentuales en TE_CIT/PIB.\n\n")
```

```
##   un cambio de 1.8142 puntos porcentuales en TE_CIT/PIB.
```

```
# Modelo 2
b1_m2 <- coef(mod2)["EleccionesLimpias"]
cat("Modelo 2 (log-lineal):\n")
```

```
## Modelo 2 (log-lineal):
```

```
cat("  β₁ =", round(b1_m2, 4), "\n")
```

```
##   β₁ = 11.1204
```

```
cat("  Interpretación: Un aumento de 1 punto en EleccionesLimpias se asocia con\n")
```

```
##   Interpretación: Un aumento de 1 punto en EleccionesLimpias se asocia con
```

```
cat("  un cambio de", round(b1_m2 * 100, 2), "% en TE_CIT/PIB.\n")
```

```
##   un cambio de 1112.04 % en TE_CIT/PIB.
```

```
cat("  (β₁ × 100 =", round(b1_m2 * 100, 2), "%)\n\n")
```

```
##   (β₁ × 100 = 1112.04 %)
```

```
# signo
if (b1_m2 < 0) {
  cat("  → signo NEGATIVO: mayor democracia → menor gasto tributario CIT.\n")
  cat("    Consistente con H1 del TFM: el escrutinio democrático\n")
  cat("    reduce la captura regulatoria de las empresas.\n")
} else {
  cat("  → signo POsITIVO: mayor democracia → mayor gasto tributario CIT.\n")
    cat("    Podría reflejar que democracias más maduras tienen sistemas\n")
  cat("    fiscales más desarrollados con mayor número de beneficios formalizados.\n")
}
```

```
##   → signo POsITIVO: mayor democracia → mayor gasto tributario CIT.
##     Podría reflejar que democracias más maduras tienen sistemas
##     fiscales más desarrollados con mayor número de beneficios formalizados.
```

Información de sesión

```
sessionInfo()
```

```
## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 22631)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United Kingdom.utf8
## [2] LC_CTYPE=English_United Kingdom.utf8
## [3] LC_MONETARY=English_United Kingdom.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United Kingdom.utf8
##
## time zone: Europe/Madrid
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] corrplot_0.95   ggplot2_3.5.2   sandwich_3.1-1  tseries_0.10-59
##  [5] lmtest_0.9-40   zoo_1.8-15      car_3.1-5       carData_3.0-6
##  [9] fBasics_4052.98 moments_0.14.1  stargazer_5.2.3 dplyr_1.2.0
## [13] readxl_1.4.5    readr_2.2.0     here_1.0.2
##
## loaded via a namespace (and not attached):
##  [1] utf8_1.2.5          sass_0.4.10         generics_0.1.4
##  [4] lattice_0.22-6      hms_1.1.4           digest_0.6.37
##  [7] magrittr_2.0.3      RColorBrewer_1.1-3  evaluate_1.0.3
## [10] grid_4.5.0          fastmap_1.2.0       cellranger_1.1.0
## [13] rprojroot_2.1.1     jsonlite_2.0.0      Formula_1.2-5
## [16] scales_1.4.0        jquerylib_0.1.4     abind_1.4-8
## [19] cli_3.6.5           timeSeries_4052.112 crayon_1.5.3
## [22] rlang_1.1.7         bit64_4.6.0-1       withr_3.0.2
## [25] cachem_1.1.0        parallel_4.5.0      tools_4.5.0
## [28] tzdb_0.5.0          curl_7.0.0          vctrs_0.7.1
## [31] R6_2.6.1            lifecycle_1.0.5     bit_4.6.0
## [34] vroom_1.7.0         pkgconfig_2.0.3     gtable_0.3.6
## [37] pillar_1.10.2       bslib_0.9.0         glue_1.8.0
## [40] quantmod_0.4.28     xfun_0.52           tibble_3.2.1
## [43] tidyselect_1.2.1    rstudioapi_0.18.0   knitr_1.50
## [46] farver_2.1.2        spatial_7.3-18      htmltools_0.5.8.1
## [49] rmarkdown_2.29      xts_0.14.1          timeDate_4052.112
## [52] compiler_4.5.0      quadprog_1.5-8      TTR_0.24.4
```